

Module 7: Classssification

Part 2: LDA & QDA

Wenting Xu

Iowa State University

Recap

$$Y = \begin{cases} 0 \\ 1 \end{cases}$$

Logistic Regression: $P(Y=1 | X_1, X_2, \dots, X_p)$
→ estimate β using MLE

When does logistic regression break down?

- When there is substantial separation between the two classes, parameter estimates for the logistic regression model are unstable.
 - In some cases, when there is perfect separation, the estimates cannot converge.



- If the distribution of the predictors X is approximately normal in each of the classes and the sample size is small, then other models may be more accurate than logistic regression.

- When $p > n$ (for example, in image classification).

↗ predictor

↖ sample size

Bayes classifier

Bayes rule:
$$\Pr(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}.$$

- A **Bayes classifier** is a rule that assigns an observation x to the class with largest $\pi_k f_k(x)$.
- The problem is that it is difficult to estimate f_k . 
- Three classifiers are suggested to approximate the Bayes classifier with different estimates of f_k :
 - linear discriminant analysis (LDA)
 - quadratic discriminant analysis (QDA)
 - naive Bayes

Linear Discriminant Analysis (LDA) when $p = 1$

We use Bayes Theorem to approximate $P(Y = k | X_1)$, $k = 1, \dots, K$.

$$P(Y = k | X_1) = \frac{P(X_1 | Y = k)P(Y = k)}{P(X_1)}$$

$C=1$, pass
 $C=0$, fail

↳ posterior = prob after seeing study hours
↳ prob of passing given 8 hours

↳ prior = prob of class before seeing evidence (study hours)
↳ 70% pass

↳ how likely the study hours are within a class
↳ 8 hours is common, 0 is rare

LDA (p = 1)

- We assume $P(X_1 | Y = k)$ is a **normal distribution**.
- Normal distribution's pdf: $\frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma_k^2} (x - \mu_k)^2\right)$
 - μ_k is the true mean for the k th class.
 - σ_k^2 is the true variance for the k th class.
- We make an additional assumption that $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$. We denote this common variance as σ^2 .



LDA ($p = 1$)

$$P(Y = k | X_1) = \frac{P(X_1 | Y = k)P(Y = k)}{P(X_1)}$$

Skipping technical details. If you

- Plug normal distribution into $P(X_1 | Y_k)$.
- Denote $P(Y = k) = \pi_k$.
- Take log and rearrange terms.

For each class, we compute a score $\delta_k(x)$, and assign the observation to the class with the **highest** score.

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

LDA (p = 1)

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

- δ_k is known as the linear discriminant function.
- It gets its name from the fact that its decision boundaries are **linear** in x_1 .
- μ_k and σ^2 are parameters and need to be estimated from the data.

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

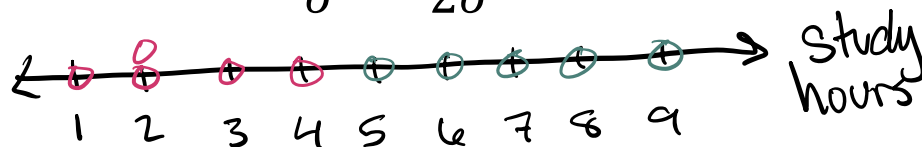
$$\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

$$\pi_k = n_k/n$$

- Compute decision boundary to assign observations to each class. The decision boundary is the “cutoff point” that separates the two classes. It’s the dividing line between different classes.

LDA Example ($p = 1$)

$$\delta_k(x_1) = x_1 \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$



$$\pi_0 = \frac{5}{10} = \frac{1}{2} \quad \hat{\mu}_0 = 2.4 \left(\frac{1+2+2+3+6}{5} \right)$$

$$\pi_1 = \frac{5}{10} = \frac{1}{2} \quad \hat{\mu}_1 = 7 \left(\frac{5+6+7+8+9}{5} \right)$$

$$\delta_0(x_1) = x_1 \cdot \frac{2.4}{5.45} - \frac{2.4^2}{2 \cdot 5.45} + \log(0.5)$$

$$\delta_1(x_1) = x_1 \cdot \frac{7}{5.45} - \frac{7^2}{2 \cdot 5.45} + \log(0.5)$$

New Student
($x_1 = 6.5$)

$\delta_0(6.5) > \delta_1(6.5)$
 $\hookrightarrow$ fail

LDA ($p > 1$)

- We assume that $X = (X_1, X_2, \dots, X_p)$ is drawn from a multivariate Gaussian (multivariate normal) distribution.
- Each class has its own mean vector μ_k (p -dimensional)
- All classes share a common covariance matrix Σ .
- Specifically, we write this as the observations in the k th class are drawn from $N(\mu_k, \Sigma)$.

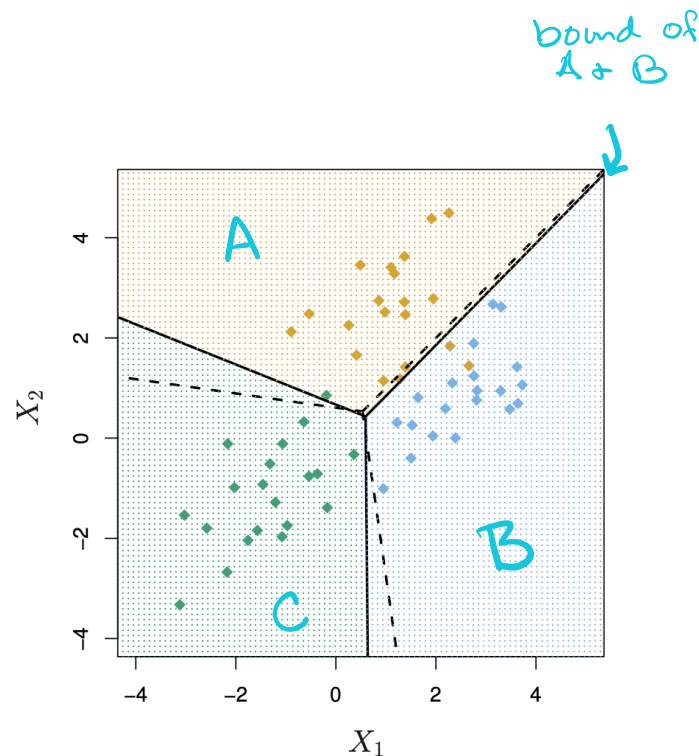
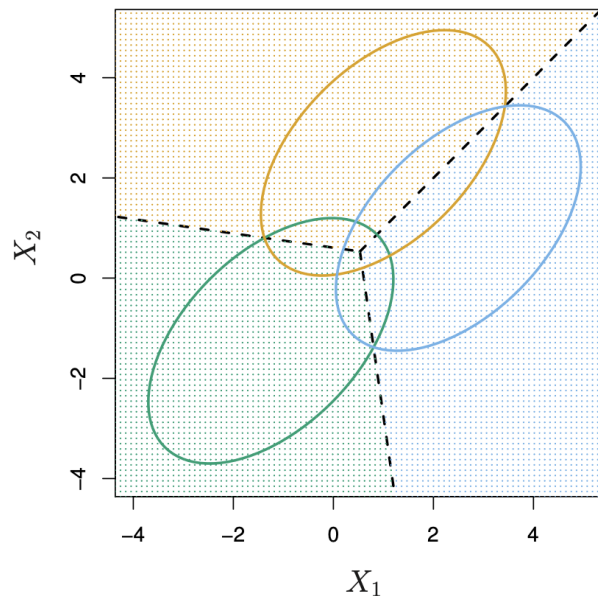
LDA for $p > 1$

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$$

- First estimate π_k , μ_k , and Σ .

Compute decision boundary to assign observations to each class.

An example with three classes



LDA: solid black line
Bayes : dashed line

Quadratic Discriminant Analysis (QDA)

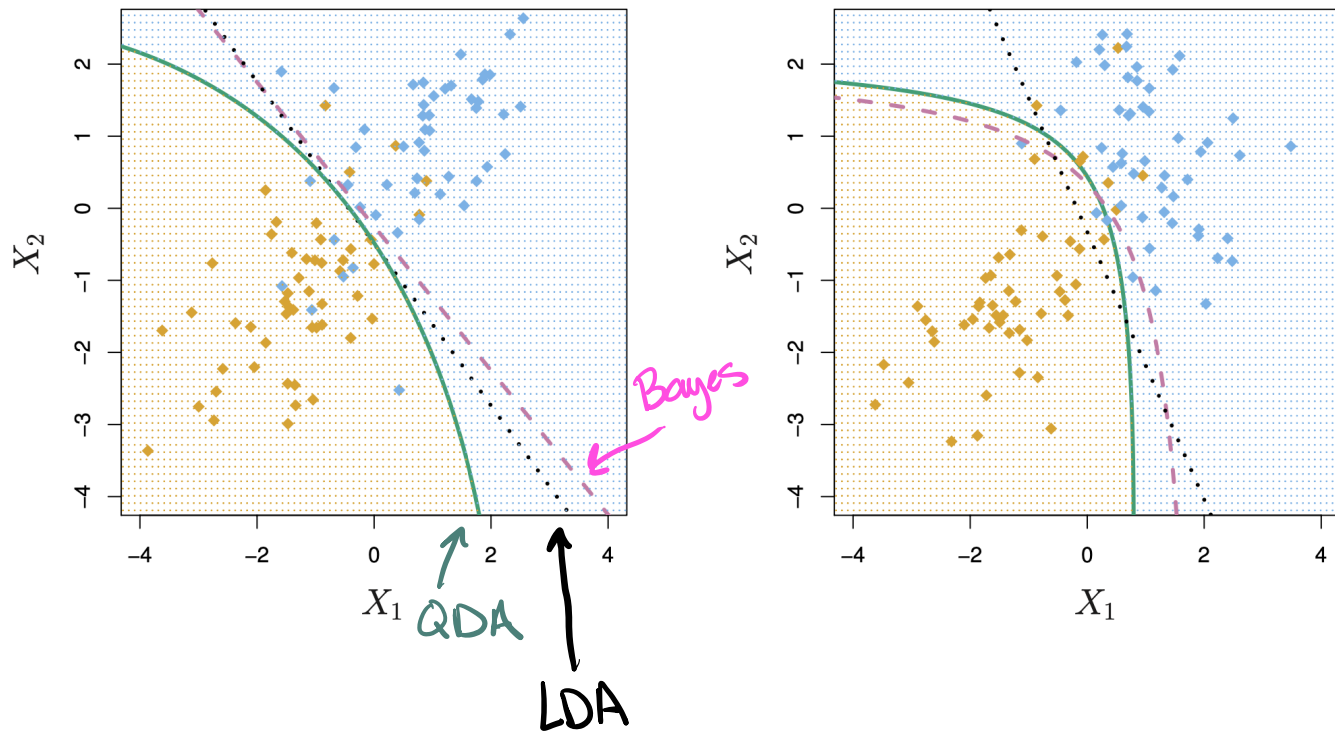
- Common covariance assumption Σ can be quite restrictive.
- QDA relaxes this assumption.
- Each class can have its own covariance matrix:

$$X \sim N(\mu_k, \Sigma_k)$$

- Σ_k is the covariance matrix for the k th class.
- Now we need to estimate both μ_k and Σ_k for each class:

$$\begin{aligned} \delta_k(x) &= -\frac{1}{2}x^T \Sigma_k^{-1}x + x^T \Sigma_k^{-1}\mu_k - \frac{1}{2}\mu_k^T \Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| \\ &\quad + \log(\pi_k) \end{aligned}$$

Quadratic decision boundaries (QDA)



Why don't we just always use QDA?

- If constant variance assumption holds, LDA dominates
- Bias-Variance Tradeoff
 - ↳ LDA = $\backslash$ less flexible; high bias, low variance
 - ↳ QDA = $\cup$ more flexible
- QDA requires more training data

R code

- See script: `lda_qda.R`